

SIM1002 Calculus I

Tutorial 6

Derivatives

Exercise 3.6

Derivative Calculations

In Exercises 1-3, given $y = f(u)$ and $u = g(x)$, find $\frac{dy}{dx} = f'(g(x))g'(x)$.

1. $y = \cos u, u = -x/3$

2. $y = \sin u, u = x - \cos x$

3. $y = -\sec u, u = \frac{1}{x} + 7x$

In Exercises 4-7, write the function in the form $y = f(u)$ and $u = g(x)$. Then find $\frac{dy}{dx}$ as a function of x .

4. $y = \left(\frac{\sqrt{x}}{2} - 1\right)^{-10}$

5. $y = \left(\frac{x^2}{8} + x - \frac{1}{x}\right)^4$

6. $y = \cot\left(\pi - \frac{1}{x}\right)$

7. $y = 5 \cos^{-4} x$

Find the derivatives of the functions in Exercises 8-14.

8. $s = \sin\left(\frac{3\pi t}{2}\right) + \cos\left(\frac{3\pi t}{2}\right)$

9. $y = \frac{1}{x} \sin^{-5} x + \frac{x}{3} \cos^3 x$

10. $y = (2x - 5)^{-1}(x^2 - 5x)^6$

11. $g(x) = \frac{\tan(3x)}{(x+7)^4}$

12. $y = \left(t^{-\frac{3}{4}} \sin t\right)^{\frac{4}{3}}$

13. $y = \sin\left(\sqrt{1 + \sqrt{t}}\right)$

$$14. \quad y = \sqrt{3t + \sqrt{2 + \sqrt{1-t}}}$$

Second derivatives

Find y'' in Exercises 15-17.

$$15. \quad y = \left(1 + \frac{1}{x}\right)^3$$

$$16. \quad y = (1 - \sqrt{x})^{-1}$$

$$17. \quad y = x^2(x^3 - 1)^5$$

For each of the following functions, solve both $f'(x) = 0$ and $f''(x) = 0$ for x .

$$18. \quad f(x) = x(x - 4)^3$$

$$19. \quad f(x) = \sec^2 x - 2 \tan x \text{ for } 0 \leq x \leq 2\pi.$$

Finding Derivative Values

In Exercises 20-26, find the value of $(f \circ g)'$ at given value of x .

$$20. \quad f(u) = u + \frac{1}{\cos^2 x}, \quad u = g(x) = \pi x, \quad x = \frac{1}{4}.$$

$$21. \quad f(u) = \left(\frac{u-1}{u-2}\right)^2, \quad u = g(x) = \frac{1}{x^2} - 1, \quad x = -1.$$

$$22. \quad \text{If } r = \sin(f(t)), f(0) = \frac{\pi}{3}, \text{ and } f'(0) = 4, \text{ then what is } \frac{dr}{dt} \text{ at } t = 0.$$

23. Suppose that the functions f and g and their derivatives with respect to x have the following values at $x = 0$ and $x = 1$.

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
0	1	1	5	$1/3$
1	3	-4	$-1/3$	$-8/3$

Find the derivatives with respect to x of the following combinations at the given value of x .

$$a) \quad g(f(x)), \quad x = 0$$

$$b) \quad f(x + g(x)), \quad x = 0$$

$$c) \quad (x^{11} + f(x))^{-2}, \quad x = 1$$

$$d) \quad \frac{f(x)}{(g(x)+1)}, \quad x = 1$$

Exercises 3.7

Differentiating implicitly

Use implicit differentiation to find $\frac{dy}{dx}$ in Exercises 1-3.

1. $x^3 + y^3 = 18xy$

2. $x^3 = \frac{2x-y}{x+3y}$

3. $xy = \cot(xy)$

Second Derivatives

In Exercises 4-6, use implicit differentiation to find $\frac{dy}{dx}$ and then $\frac{d^2y}{dx^2}$. Write the solutions in terms of x and y only.

4. $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1$

5. $xy + y^2 = 1$

6. $\sin y = x \cos y - 2$

7. If $xy + y^2 = 1$, find the value of $\frac{d^2y}{dx^2}$ at the point $(0, -1)$.

In Exercises 8-10, verify that the given point is on the curve and find the lines that are **(a)** tangent and **(b)** normal to the curve at the given point.

8. $y^2 - 2x - 4y - 1 = 0, (-2, 1)$

9. $x^2 - \sqrt{3}xy + 2y^2 = 5, (\sqrt{3}, 2)$

10. $x \sin(2y) = y \cos(2x), \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$